

Model predictive controller for path following ships validated by experimental model tests

Hongwei He^{a,*}, Thibaut Van Zwijnsvoorde^b, Evert Lataire^a, Guillaume Delefortrie^a

^a Ghent University, Ghent, Belgium

^b Flanders Hydraulics, Antwerp, Belgium

ARTICLE INFO

Keywords:

Autonomous navigation
Model predictive control
Path following
Free running model tests
Underactuated ships
Ship hydrodynamics

ABSTRACT

This paper introduces a practical model predictive controller for surface ships' path following, with validation through experimental tests. Model predictive control (MPC) is a state-of-the-art modern control technique, providing local optimal control action based on prediction and reference. In this approach, a simple predictive model based upon the Nomoto model is employed to forecast the (near) future ship positions, while the Serret-Frenet frame and the Spline path function are used to generate a smooth ship speed-based reference. Two optimizers, namely the Spline optimizer and the Cobyla (Constrained Optimization by Linear Approximation) optimizer, are used to retrieve an optimal solution that minimizes the difference between the prediction and the reference. To achieve a real-time control feedback, the optimization variables are reduced significantly to lower calculation time by assuming a constant rudder angle throughout the prediction horizon, which is consistent with a realistic human controlled manoeuvre. In addition, an adaptive prediction horizon strategy is proposed to balance overshoot and accuracy.

A series of free running model tests were conducted in the Towing Tank for Manoeuvres in Shallow Water (cooperation Flanders Hydraulics – Ghent University) in Ostend, Belgium, in order to evaluate the effectiveness and performance of the proposed controller, and to explore the impact of its parameter settings. The 20 m wide towing tank provides ample space to design complex paths for the ship to follow, which poses a significant challenge for the controller. Essential parameters in MPC, including weight factor for heading error, parameters of predictive model, prediction horizon, gain of adaptive prediction horizon and optimizer, are described in detail and their effects on the control outcomes are discussed with analysis of experimental results. The results demonstrate excellent control performance of the MPC, i.e. high tracking accuracy, smooth track transition, insensitivity to parameter setting and strong robustness in the presence of inherent model uncertainty, despite the simplifications made for practicality.

1. Introduction

Autonomous ships have emerged as a prominent trend in the shipping industry, owing to its promising prospect for enhancing safety and reducing costs. Within the realm of autonomous ship development, automatic control techniques play a pivotal role. These control techniques are commonly classified into two categories, namely model-free control and model-based control algorithms. Model-free control methods, such as PID control and Fuzzy control, do not rely on an inherent model of the controlled system. Instead, they work based on the error between the actual system states and their targets/references, resulting in error feedback control. However, model-based control

techniques employ an inherent model of the system to estimate its response to the applied control inputs. Subsequently, a control law is designed with the objective of achieving a stable error system, wherein the error of interest is asymptotically reduced over time. In scenarios where only the states at the next time step are estimated, the resulting control behaviour is limited to local feasibility due to the lack of foresight into future positions. Nonetheless, this approach demonstrates notable tolerance to model mismatch, owing to the limited requirement of predicting only one step forward. Examples of this approach include sliding-mode control (Zhang et al., 2000), backstepping control (Deng et al., 2017), and active disturbance rejection control (Zhang et al., 2022a). When the prediction horizon is extended to an infinite horizon,

* Corresponding author.

E-mail address: Hongwei.He@UGent.be (H. He).

<https://doi.org/10.1016/j.oceaneng.2023.115971>

Received 13 June 2023; Received in revised form 2 August 2023; Accepted 5 October 2023

Available online 31 October 2023

0029-8018/© 2023 Elsevier Ltd. All rights reserved.

global optimal solutions are provided but rely heavily on the prediction of the inherent model, such as Linear Quadratic Regulator (Zhao et al., 2022), thus the performance is easily impacted by model uncertainties. On the other hand, Model Predictive Control (MPC), an important branch of model-based control types, involves predicting and utilizing the states over a finite future period, known as the prediction horizon, to determine the control behaviour. It is a balance between the one-step-prediction-based control and the infinite prediction-based control and combines the merits of both. This approach provides a local optimal solution within the prediction horizon, allowing for dynamic programming to compensate for prediction deviations. A predictive model, either refined or approximate, is utilized to estimate future states. After that, the difference between the reference states and the predicted states is evaluated by a predefined cost function (the objective function), which allows for easy extension to include additional costs, such as the offset sum of control behaviours, and can be constrained by system dynamics, safety criteria, navigation rules and alike. Consequently, a conclusion is drawn that the key essentials of MPC encompass the predictive model, the reference and the optimization.

The selection of an appropriate predictive model that describes the dynamic response of a vessel to control behaviours directly impacts the output of MPC. A review of relevant recent literature, as summarized in Table 1, reveals various choices of predictive models employed in MPC-based automatic control of ship motion across different implementation scenarios. Among these options, delicate nonlinear models, simplified nonlinear models and linearized models are commonly used, whilst black-box models and the Nomoto model are also candidates. Delicate nonlinear models, while providing higher prediction accuracy, often entail a larger number of model parameters that need to be determined in practice, thereby adding complexity to controller implementation. To mitigate this challenge, delicate nonlinear models can be simplified by ignoring high-order and insignificant terms or linearized at a specific operating point, e.g. the normal/latest operating point or the equilibrium point, to reduce complexity or facilitate the implementation of a linear MPC. The resulting simplified or linearized models typically involve fewer model parameters. Alternatively, a very simple course model for the ship's manoeuvring behaviour (Papadimitrakakis et al., 2021) employing the Nomoto model to estimate course change and kinetic equations to calculate positions under constant speed assumptions requires the fewest number of model parameters. However, it is

important to note that such course model is applicable only to path following or heading control scenarios. The determination of the model parameters is a big handicap in implementing MPC. Most simulation studies adopt benchmark or prototype vessel models whose hydrodynamic coefficients are provided through captive tests, and then employs them as the predictive model directly (after simplification or linearization) which closes the loop between output and input resulting in highly accurate mathematical models. However, in real life, it is impractical/impossible to perform captive tests for every ship to be controlled. Other shortcuts should be investigated. The data-driven method, system identification, offer promising alternatives for modelling a vessel's hydrodynamics based on systematic input-output data, yielding nonparametric (black-box) or parametric predictive models. Furthermore, online system identification techniques can account for model uncertainty due to external disturbances and changes in operating conditions. Additionally, certain software tools, such as Vessel Response, can estimate added mass and damping matrices based on vessel hull geometry, providing an acceptable solution.

It is important to emphasize that the result cannot be convincing if the predictive model and the simulation model is almost the same in simulation studies where the inherent model is known and well provided with detailed hydrodynamic coefficients. The robustness of such controllers to the uncertainty of the predictive model cannot be validated.

Aiming to maintain good practicality, the present paper employs a course model that requires very few parameters as the predictive model for path following. The course model is modified with a bias term and is identified with data collected from zigzag manoeuvres, as opposed to the empirical estimation as done in (Papadimitrakakis et al., 2021). The identified model effectively captures the yaw dynamics, which is crucial in path following, thereby enhancing control performance.

The reference signal serves as the target for the control system. Automatic control performance is limited if the reference is not well generated. For a path following problem where the ship is steered to a time-independent desired path, the design of the controller aims to reduce the cross track error, and line-of-sight (LOS) guidance is extensively employed to convert it into a heading control problem, which is particularly preferable for underactuated ships for which some degrees of freedom are not controllable.

Table 2 demonstrates that LOS can also be implemented in MPC for generating the reference heading. However, since the LOS angle is

Table 1

Predictive models adopted in MPC-based automatic control of ship motion, in the relevant literature within the past five years.

Literature	Model type	Number of parameters	Method to determine parameters	Difference to control plant	Implementation
Xu et al. (2023)	BBM	Nonparametric	Offline SI	Big difference	WF
Zhang et al. (2022b)	LM_LOP	>10	Provided in other literature, captive tests	Some difference	TT & CA
Sandeepkumar et al. (2022)	SNM	>10	Provided in other literature, captive tests	Some difference	PF
Martinsen et al. (2022)	DNM	>10	Online SI	Big difference especially in sea trials	TT
Xue et al. (2021)	LM_NOP_F	5–10	Offline SI	Big difference	TT & CA
Papadimitrakakis et al. (2021)	CM	<5	Empirical guess	Big difference	CA
Zheng et al. (2020)	LM_EQ_F	>10	Provided in other literature, captive tests	Some difference	DP
Li et al. (2020b)	DNM	>10	Provided in other literature, captive tests	No difference	AD
Li et al. (2020a)	DNM	>10t	Provided in other literature, captive tests	No difference	PF
Shen et al. (2019)	SNM	>10	Provided in other literature, offline SI	Big difference	PF
Putri et al. (2019)	LM_NOP	5–10	Provided in other literature, not clear	Some difference	PF & CA
Abdelaal et al. (2018)	SNM	5–10	Calculated by using the Vessel Response software	Not clear that the same model was used for simulation or not	TT & CA

*note: WF = waypoint following, TT = trajectory tracking, CA = collision avoidance, PF = path following, DP = dynamic positioning, AD = automatic docking; BBM = black-box model, SNM = simplified nonlinear model, DNM = delicate nonlinear model, CM = course model based on the Nomoto model, LM_LOP = linearized model at the latest operating point, LM_EQ_F = linearized model at the equilibrium point with forces as inputs, LM_NOP = linearized model at the normal operating point, LM_NOP_F = linearized model at the normal operating point with forces as inputs, SI = system identification.

Table 2

Selection of LOS mechanisms adopted to generate reference, in the relevant literature about MPC-based path following.

Literature	Reference	Description	Implementation
Xu et al. (2023)	LOS angle (outdated varying track error)	$\psi_{LOS} = \alpha_k - \tan\left(\frac{e_k}{\Delta}\right)$, $e_k = f(x_k, y_k, x_k^{ref}, y_k^{ref})$	PF
Sandeepkumar et al. (2022)	LOS angle (outdated constant track error)	$\psi_{LOS} = \alpha_k - \tan\left(\frac{e}{\Delta}\right) - \beta$	PF
Li et al. (2020c)	LOS angle (outdated constant track error)	$\psi_{LOS} = \alpha_k - \tan\left(\frac{e}{\Delta}\right)$	PF
Oh and Sun (2010)	LOS angle (varying track error)	$\psi_{LOS} = \alpha_k - K_k e_k$, $e_k = f(x_k, y_k, x_k^{ref}, y_k^{ref})$	PF

*note: ψ_{LOS} is the LOS reference angle, α_k , e_k , x_k^{ref} , y_k^{ref} and K_k are the path direction, cross track error, reference x-position, reference y-position and LOS gain at the k -th prediction step respectively, β is the drift angle and f is the function to calculate cross track error.

calculated based on the cross track error, position prediction is necessary prior to generating the reference, resulting in varying reference series for different control behaviour candidates. This further complicates the optimization problem. In some studies, simplifications have been made, such as using the latest measured ship position to estimate all future cross track errors within each prediction horizon as demonstrated in (Xu et al., 2023), or simply utilizing the latest real track error as shown in (Li et al., 2020c; Sandeepkumar et al., 2022).

Table 3 presents alternative methods to provide reference purely based on the path itself in MPC. Among them, (Shen et al., 2019) introduces a clear and comprehensive approach wherein a smooth parametric path function is employed to represent the path, enabling efficient computation of path-related information, e.g. position, slope and curvature. In this study, a similar methodology is adopted, but augmented with the Serret-Frenet frame and generation of the parametric path function based on waypoints.

Efficiency in solving the formulated MPC optimization problem is of paramount importance for real-time control. Specifically, the calculation time consumed in obtaining the next control behaviour(s) must be within the control horizon (in time unit). MPC generates optimal control actions for each time step within a prediction horizon. The ship has the option to either follow all of these instructions sequentially within the given period or selectively consider only the initial upcoming instructions and make updates once they have been executed. The length of selected time steps is namely the control horizon. Typically, the control horizon is one time step to improve the controller's robustness against errors from the predictive model and uncertainties from the control plant (e.g. environmental disturbance and varying ship dynamics due to the changed loading condition or water depth), therefore normally the calculation time should be smaller than the sampling time. In addition, fast computation allows for higher sampling frequency that enables more frequent updates of ship states, leading to stronger controller's robustness in the face of prediction uncertainty. Table 4 shows the calculation time per time step of various MPC algorithms reported in the literature. It worth noting that the calculation time is not really comparable due to the different in computer performance and prediction horizon, but the readers can get a brief idea. Most of them are quite efficient and performed in Matlab. However, in the present case, given the requirement for a C# environment and a fixed sampling frequency of 8 Hz in the free-running system, the MPC optimization problem is significantly simplified to boost efficiency. To this end, two derivative-free optimizers are adopted, one of which is an open-source solver for constrained nonlinear optimization problems, while the other is a custom-designed optimizer similar to the scenario-based MPC. The scenario-based MPC (Akdağ et al., 2022; Hagen et al., 2022; Johansen et al., 2016) is suitable for collision avoidance and selects the optimal solution from a small predefined set of representative candidates (control behaviours: course offsets and speed commands) that remain constant in a prediction horizon, resulting in computational efficiency. In order to maintain high accuracy in path following, after the evaluation on the predefined set, the corresponding cost set is fitted and

interpolated to identify a more suitable solution.

Most studies in the literature primarily present simulation results, whereas it is important to show validation by experiments. Table 5 provides an overview of the literature that incorporates experimental tests. In this paper, the focus is on the validation of the proposed MPC method for path following with scale model experimental tests conducted in an controllable towing tank environment. A series of tests are performed and rich demos are shown to elucidate the effects of each parameter setting and the robustness of the designed controller. Control performance is quantitatively evaluated and can be referred to in future similar studies. The findings substantiate that based on a very simple identified predictive model and a few tuneable parameters, MPC can be successfully implemented in real-time path following of a physical ship

Table 3

Reference generated purely based on the path, in the literature about MPC-based path following, to name but a few.

Literature	Reference	Description	Sampling method	Implementation
Li et al. (2020a)	Straight-line path (path direction = 0)	$p_y(\varpi) = 0$	Not necessary	PF
Shen et al. (2019)	Smooth parametric path, $p_x(\varpi)$, $p_y(\varpi)$	$p_x(\varpi)$, $p_y(\varpi)$, $p_\psi(\varpi) = \text{atan } 2(p'_y, p'_x)$, $p_u(\varpi) = \sqrt{p_x'^2 + p_y'^2} \varpi$, $p_v(\varpi) = 0$, $p_r(\varpi) = \frac{p'_x p_y - p_x p'_y}{p_x'^2 + p_y'^2} \varpi$	Using an integrator to describe the path dynamics: $\dot{\varpi} = v_\varpi$ (v_ϖ is the desired speed of path variable)	PF
Putri et al. (2019)	Dubins path	$p_x(\varpi) = 0$, $p_r(\varpi) = 0$, $p_\psi(\varpi) = \text{atan } 2(p'_y, p'_x)$, $p_e(\varpi) = 0$	Not clear	PF & CA
Ghaemi et al. (2010)	Straight-line path (path direction = 0)	$p_e(\varpi) = 0$, $p_\psi(\varpi) = 0$	Not necessary	PF
Li et al. (2009)	Smooth path, Serret-Frenet frame	$p_e(\varpi) = 0$, $p_\psi(\varpi) = \text{atan } 2(p'_y, p'_x)$	Not necessary in the simulation demos (a straight-line path was to refer to)	PF

*Note: Sampling method means the way the reference points are sampled within a prediction horizon. ϖ is a path variable to locate a point on the path. p_x , p_y , p_ψ , p_u , p_v , p_r and p_e represent the reference x-position, y-position, heading, surge velocity, sway velocity, yaw rate and cross track error respectively.

Table 4

Calculation efficiency of different MPC optimizers reported in the relevant literature, to name but a few.

Literature	Optimizer	MPC type	Average calculation time [s]	Max calculation time [s]	Sampling time [s]	Prediction horizon/step	Implementation	Computer environment
Xu et al. (2023)	LMI	CMPC/ SMPC	0.11	0.3	1	15	PF	–
Li et al. (2020b)	fmincon-ip	LMO- NMPC	0.25–0.4	5.05 (rare)	1	30	AD	Env A
Abdelaal et al. (2018)	ACADO- GNI	NMPC	–	0.004	5	40	TT & CA	Env B
Shen et al. (2017)	C/GMRES	NMPC	0.01	–	0.1	20	TT	Env C
Zheng et al. (2014)	fmincon	LMPC	0.14	–	0.5	10	TT	Env D
Guerreiro et al. (2014)	WLS	NMPC	0.01	0.06	0.2	50	TT	Env E
Oh and Sun (2010)	SQP	LMPC	0.02	–	0.5	10	PF	Env F
Ghaemi et al. (2010)	InPA-SQP	LMPC/ NMPC	–	<0.01	0.01	7	PF	Env G

*Note: **Optimizers:** LMI = LMI (Linear Matrix Inequalities) based optimizer; fmincon-ip = Matlab fmincon function, interior-point method; ACADO- GNI = ACADO toolkit (C-code), Gauss-Newton iterative algorithm; C/GMRES = Continuation/Generalized Minimal RESidual; WLS = Wolfe line search; SQP = standard QP algorithm; InPA-SQP = Integrated Perturbation Analysis and Sequential Quadratic Programming.

MPC types: SMPC = Stochastic MPC; CMPC = Cautious MPC; NMPC = Nonlinear MPC; LMPC = Linear MPC; LMO-NMPC = Lexicographic Multi-Objective NMPC.

Computer environments: Env A = Intel Core i7-7500, 8 GB RAM, Matlab R2019a; Env B = 3.3 GHz Core i5, 8 GB RAM, Matlab 2014b; Env C = 2.90 GHz Intel Core i7-3520 M, 4 GB RAM; Env D = Matlab 2011b; Env E = Intel Core i7–2640 M, Matlab; Env F = 2.2 GHz Pentium 4 PC, 2G RAM; Env G = PC-based PCI04 hardware, QNX real-time operating system.

model with high accuracy.

2. Guidance system

2.1. Path generation

The reference path is generated based on waypoints, but unlike connecting all waypoints by straight line segments, a cubic spline is employed to define a smooth path where both the first and second derivatives are continuous. This guarantees that the curvature, critical path parameter in curve tracking, remains continuous along the path. The cubic spline utilizes piecewise third-order polynomials to pass through a set of waypoints:

$$\begin{aligned} p_x(\varpi) &= a_3\varpi^3 + a_2\varpi^2 + a_1\varpi + a_0 \\ p_y(\varpi) &= b_3\varpi^3 + b_2\varpi^2 + b_1\varpi + b_0 \end{aligned} \quad (2.1)$$

where a_i and b_i ($i = 0, 1, 2, 3$) are polynomial coefficients determined by the constraints imposed by the first and second-order derivatives. ϖ is set to the waypoint indexes in this paper.

The partial derivatives of $p_x(\varpi)$ and $p_y(\varpi)$ with respect to ϖ are

$$\begin{aligned} p'_x(\varpi) &= 3a_3\varpi^2 + 2a_2\varpi + a_1 \\ p'_y(\varpi) &= 3b_3\varpi^2 + 2b_2\varpi + b_1 \end{aligned} \quad (2.2)$$

Hence, the desired ship speed $U_d(t)$ can be computed as

$$\begin{aligned} \dot{p}_x(t) &= p'_x(\varpi)\dot{\varpi} \\ \dot{p}_y(t) &= p'_y(\varpi)\dot{\varpi} \\ U_d(t) &= \sqrt{\dot{p}_x(t)^2 + \dot{p}_y(t)^2} = \sqrt{p'_x(\varpi)^2 + p'_y(\varpi)^2} \dot{\varpi} \end{aligned} \quad (2.3)$$

The curvature κ is calculated based on its definition as:

$$\kappa(\varpi) = \frac{|p'_x(\varpi)p''_y(\varpi) - p''_x(\var�)p'_y(\varpi)|}{(p'_x(\varpi)^2 + p'_y(\varpi)^2)^{3/2}} \quad (2.4)$$

2.2. Serret-Frenet frame

The ship position is projected onto the Serret-Frenet (SF) frame (Frenet, 1852), which is a path-tangential reference frame moving along

the planned path. As shown in Fig. 1, U is the ship speed, and (s, e) , χ_{SF} , ψ_{SF} are the relative position, course and heading of the ship in the SF frame respectively. The $\{n\}$, $\{b\}$ and $\{SF\}$ notations represent the earth-fixed, body-fixed and SF frames respectively. When $s = 0$, e can be regarded as the closest lateral distance from the ship to the planned path. Consequently, s becomes an additional variable to control, which can be achieved by managing the movement of the SF frame.

Fossen (2011) introduced a speed dynamic equation for the SF frame, which enables it to closely track the motion of the ship:

$$\dot{\varpi} = \frac{U \cos(\chi_{SF}) + K_1 s}{\sqrt{p'_x(\varpi)^2 + p'_y(\varpi)^2}} \quad (2.5)$$

where K_1 is tunable positive parameter that determines the convergence speed of s .

2.3. Reference generation

The cubic spline function describes the planned path, from which N_p (prediction horizon) points are sampled to generate the reference for MPC within each prediction horizon. Considering that a constant ship speed is assumed in path following, based on Eq. (2.3), the path variable ϖ is integrated as

$$\varpi_{i+1} = \varpi_i + \frac{U_d h}{\sqrt{p'_x(\varpi)^2 + p'_y(\varpi)^2}}, i = 1, \dots, N_p \quad (2.6)$$

Then, the path information at each point defines the references:

$$\begin{aligned} p_x(\varpi_i) &= a_3\varpi_i^3 + a_2\varpi_i^2 + a_1\varpi_i + a_0 \\ p_y(\varpi_i) &= b_3\varpi_i^3 + b_2\varpi_i^2 + b_1\varpi_i + b_0 \\ p_\chi(\varpi_i) &= \text{atan} 2(p'_y(\varpi_i), p'_x(\varpi_i)) \end{aligned} \quad (2.7)$$

where p_χ represents the reference ship course and is obtained as the tangent direction. If required, the reference surge and sway velocity and yaw rate can be set accordingly:

Table 5

Relevant literature presenting experimental test results for MPC-based automatic control of ship motion, to name but a few.

Literature	Physical model, dimension, sampling frequency, test environment, implementation	Strength	Weakness
Martinsen et al. (2022)	milliAmpere (urban passenger ferry), 5 m × 2.8 m, 5 Hz, outdoor environment, TT	Sea trails; online system identification; online tuning of cost function parameters; comparison between simulation and experiment; comparison with another controller	Potential problematic parameter updates; more computationally demanding than more traditional methods; complex algorithms; no discussion about other MPC parameter settings; adopting forces as control inputs (not applicable for all full-scale ships)
Wang et al. (2018)	3D-printing cube-like robo-boat, 0.9 m × 0.45 m, 5 Hz, swimming pool & river, TT	Indoor and outdoor tests; system identification of the model; efficient optimization; comparison between simulation and experiment	Limited tracking accuracy: 0.72 (indoor), 0.42 (outdoor); no discussion about parameter setting; Adopting forces as control inputs
Zhang et al. (2017)	Ship model, 0.3 m, swimming pool, PF	Disturbance compensation: environmental disturbance (wind from a fan); roll constraint; comparison with another study	Poor demos; unclear reference; no quantitative evaluation; no explanation to parameter setting
Ghaemi et al. (2010)	Ship model, 1.68 m × 0.38 m, 100 Hz, towing tank, PF	Propeller speed could be controlled; efficient optimization	Simple application (following a straight line); poor demo; no quantitative evaluation; no explanation to parameter setting

*Note: tracking accuracy = 1 – average tracking error/half of ship width.

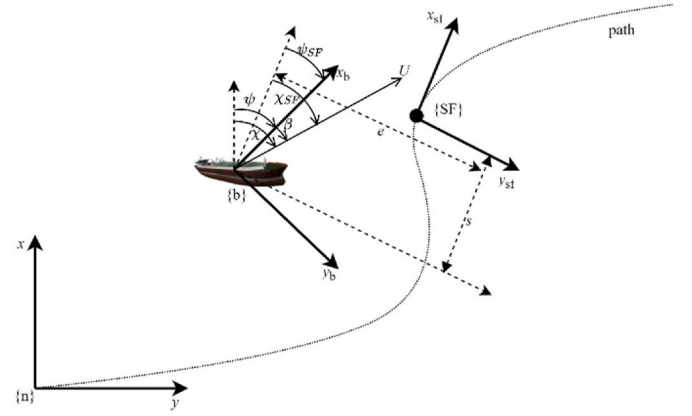
$$\begin{aligned}
 p_u(\varpi_i) &= U_d \\
 p_v(\varpi_i) &= 0 \\
 p_r(\varpi_i) &= \dot{p}_\psi(\varpi) = \frac{p_x' p_y'' - p_x'' p_y'}{p_x' + p_y'} \varpi
 \end{aligned} \quad (2.8)$$

where the reference yaw rate is obtained by differentiating p_ψ . In path following, where the ship maintains a constant propeller rate, U_d is then set to the latest measured ship speed.

3. Model predictive control

3.1. Outline

Fig. 2 illustrates the general framework of MPC. At each time step, it looks ahead to the near future through the predictive model, which enables the system to take action in advance and search for optimal control actions with multiple objectives and constraints. The easily extendable objective function and constraints facilitate a custom design, which for instance, can include reducing tracking error, avoiding obstacles and saving energy. The real-time update of the optimization problem tolerates the prediction error and other uncertainties, a characteristic known as dynamic programming. Dynamic programming

**Fig. 1.** Serret-Frenet frame.

enhances robustness and improves control accuracy in the presence of uncertainties.

3.2. Optimization problem

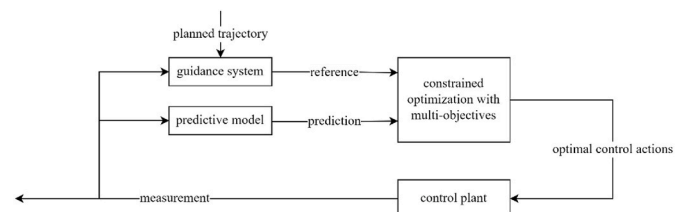
The fundamental concept of MPC involves solving an optimization problem defined by an objective function while subjected to several constraints. For multi-objective optimization, the objective function can be the addition of several weighted sub-objectives. In the case of trajectory tracking and collision avoidance, MPC is expected to reduce the difference between the predicted and reference trajectory while minimizing actuator efforts and avoiding collisions with detected obstacles. Accordingly, the optimization problem can be formulated as follows:

$$\text{argmin} J(\mathbf{x}, \mathbf{u}) = \sum_{i=1}^{N_p} (\mathbf{x}_i - \mathbf{x}_i^{\text{ref}})^T \mathbf{Q} (\mathbf{x}_i - \mathbf{x}_i^{\text{ref}}) + \Delta \mathbf{u}_{i-1}^T \mathbf{R} \Delta \mathbf{u}_{i-1}$$

subject to:

$$\begin{aligned}
 \mathbf{x}_{i+1} &= \mathbf{x}_i + f(\mathbf{x}_i, \mathbf{u}_i)h; \\
 \mathbf{u}_{\min} &\leq \mathbf{u}_i \leq \mathbf{u}_{\max} \\
 \Delta \mathbf{u}_{\min} &\leq \Delta \mathbf{u}_i \leq \Delta \mathbf{u}_{\max} \\
 (x_i - x_{j,i}^{\text{obs}})^2 + (y_i - y_{j,i}^{\text{obs}})^2 &\geq D_{j,i}^2; \quad i = 1, \dots, N_p; j = 1, \dots, N_{\text{obs}}
 \end{aligned} \quad (3.1)$$

where $J(\mathbf{x}, \mathbf{u})$ is the objective function (or cost function), $\mathbf{Q}$ and $\mathbf{R}$ are weight matrixes, $f(\mathbf{x}, \mathbf{u})$ is a nonlinear vector function describing the dynamic response of ship manoeuvring motion, h is the sampling time, and at prediction step i , $\mathbf{x}_i$ is the predicted ship state vector, $\mathbf{x}_i^{\text{ref}}$ is the reference ship state vector, $\mathbf{u}_i$ is the input vector, $\Delta \mathbf{u}_i$ is the input increment vector, x_i and y_i represent the predicted x-position and y-position of the ship, $x_{j,i}^{\text{obs}}$ and $y_{j,i}^{\text{obs}}$ represent the predicted x-position and y-position of the obstacle j , and $D_{j,i}$ is the safe distance between the own ship and the obstacle j . The ship state vectors and the input vector are composed of as follows:

**Fig. 2.** Framework of MPC

$$\begin{aligned} \mathbf{x}_i &= [x_i, y_i, \chi_i, u_i, v_i, r_i]^T \\ \mathbf{x}_i^{\text{ref}} &= [p_x(\varpi_i), p_y(\varpi_i), p_\chi(\varpi_i), p_u(\varpi_i), p_v(\varpi_i), p_r(\varpi_i)]^T \\ \mathbf{u}_i &= [\delta_i, n_i]^T \end{aligned} \quad (3.2)$$

where χ_i , u_i , v_i and r_i denote the predicted ship course, surge velocity, sway velocity and yaw rate respectively, and δ_i and n_i are the rudder angle and propeller rate. The optimal solution to the MPC problem yields the optimal control actions for the next N_p time steps. However with dynamic programming, only the optimal control actions at the first upcoming N_c steps are implemented. N_c is called control horizon, and it is usually set to 1 as also done in this study.

N_p is a crucial parameter in MPC. A larger N_p allows the controller to look further ahead into the future and leads to a more moderate reaction from the controller, but it can also result in less accurate predictions or even cause control failures when set to a really large value. On the other hand, a smaller N_p results in more accurate predictions but more aggressive control. Normally, its appropriate value ranges widely. So that N_p can be easily tuned to get satisfactory control performance.

Additionally, N_p also affects the calculation time required to solve the optimization. In the optimization problem represented by Eq. (3.1) and (3.2), $2N_p$ variables need to be optimized. The large size results in a higher computational time cost. For real-time control, optimization efficiency is required. Reducing the problem size is needed in most cases. Drawing inspiration from manual ship steering, where the helmsman typically adjusts the rudder angle after a period of time, the rudder angle and propeller rate can be treated as a constant within a prediction horizon:

$$\mathbf{u}_i = [\delta_1, n_1]^T, i = 1, \dots, N_p \quad (3.3)$$

This dramatically reduces the size of the optimization problem, with the number of optimization variables dropping to 2, and consequently reduces the calculation time. In path following, the propeller rate and velocities are not controlled. Therefore, the input vector becomes a scalar:

$$\mathbf{u}_i = \delta, i = 1, \dots, N_p \quad (3.4)$$

As a result, the constraint on the rudder angle increment can be excluded from the optimization problem. Furthermore, the paper focusses on the scenarios which have been tested in the physical towing tank, where no (dynamic) obstacles were present. However, in the future, with the aid of a carriage, it may be possible to consider ship-ship interaction by controlling another ship to meet or overtake the free-sailing ship. Finally, the optimization problem is simplified to:

$$\text{argmin}_{\delta} J(\delta) = \sum_{i=1}^{N_p} (\mathbf{x}_i - \mathbf{x}_i^{\text{ref}})^T \mathbf{Q} (\mathbf{x}_i - \mathbf{x}_i^{\text{ref}}) \quad (3.5)$$

subject to:

$$\delta_{\min} \leq \delta \leq \delta_{\max}$$

where $\mathbf{x}_{i+1} = \mathbf{x}_i + f(\mathbf{x}_i, \delta_1)h$, $i = 1, \dots, N_p$. The equality constraint is moved into the cost function to simplify the constraints.

3.3. Predictive state

Given that the optimization problem only involves optimizing the rudder angle, the yaw dynamics become dominant. To simplify the predictive model, a modified Nomoto model is utilized. This allows for the prediction of ship states using the following equation, with the assumption that the angular velocity of drift is ignorable and the ship speed U remains constant within a prediction horizon:

$$\begin{cases} \dot{\chi} = r + \dot{\beta} \approx r \\ \dot{r} = -\frac{r}{T} + \frac{K}{T} \delta - \frac{B}{T} \\ \dot{x} = U \cos \chi \\ \dot{y} = U \sin \chi \end{cases} \quad (3.6)$$

where χ is the ship course, K , T and B are respectively the gain, the time constant and the bias term in the modified Nomoto model. When B is set to 0, the modified Nomoto model reverts to the conventional one. Although this model definitely introduces significant prediction errors, dynamic programming allows for continuous updates of the prediction at each time step, enlarging tolerance to prediction errors. Its discrete form is as follows:

$$\begin{cases} x_{i+1} = x_i + Uh \cos \chi_i \\ y_{i+1} = y_i + Uh \sin \chi_i \\ \chi_{i+1} \approx \chi_i + r_i h \\ r_{i+1} = r_i - \frac{r_i h}{T} + \frac{Kh}{T} \delta - \frac{Bh}{T} \end{cases} \quad (3.7)$$

for $i = 1, \dots, N_p$.

3.4. Optimization solver

3.4.1. Spline optimizer

The Spline optimizer is designed for single-variable optimization problems currently and utilizes Spline regression and interpolation. To determine the optimal solution, the procedure involves:

1. Obtaining a cost set Ω by sampling a small number (M_s) of points of the cost function within the feasible region,

$$\begin{aligned} \Omega = \{ & (\delta_1, J(\delta_1)), (\delta_1 + \Delta\delta, J(\delta_1 + \Delta\delta)), \dots, (\delta_1 \\ & + (M_s - 2)\Delta\delta, J(\delta_1 \\ & + (M_s - 2)\Delta\delta)), (\delta_u, J(\delta_u)) \} \end{aligned}$$

$$\Delta\delta = (\delta_u - \delta_1) / M_s$$

where δ_1 and δ_u are the lower bound and upper bound of the feasible region respectively

2. Employing a cubic spline to fit the samples in Ω to obtain a surrogate function of the objective function (regression)

$$J(\delta) \approx S(\Omega, \delta)$$

3. Sampling a large number (M_l) of points of the surrogate function (interpolation)

$$\begin{aligned} \Phi = \{ & (\delta_1, J(\delta_1)), (\delta_1 + \Delta\delta, J(\delta_1 + \Delta\delta)), \dots, (\delta_1 + (M_l - 2)\Delta\delta, J(\delta_1 \\ & + (M_l - 2)\Delta\delta)), (\delta_u, J(\delta_u)) \} \end{aligned}$$

$$\Delta\delta = (\delta_u - \delta_1) / M_l$$

4. Selecting the point with the minimum cost as the optimal solution

$$\text{argmin}_{\delta \in \Phi} J(\delta)$$

The surrogate model (spline function) is used because the objective function is expensive to evaluate (time-consuming), while calculating the output of the surrogate model only is quite fast. In addition, a parameter, tolerance of cost reduction (CRT), is incorporated to decrease the frequency of changing the rudder angle. This means that if the cost reduction falls within the tolerance, the present rudder angle will not be changed. The Spline optimizer is inspired by Bayesian

Optimization (Mockus, 1989), and it can be improved through iterative optimization and extended to solve multi-variable optimization problems by equipping it with a high-dimension regressor.

3.4.2. Cobyta optimizer

Cobyta (Constrained Optimization by Linear Approximation) is a derivative-free optimization algorithm (Powell, 1994), which is able to solve a nonlinear constrained optimization problem with a moderate number of variables (around 100). It uses linear approximations with respect to the objective and constraint function, thus categorized to the sequential trust-region algorithm. The optimization problem it solves has the following general form:

$$\begin{aligned} \min F(x) \\ \text{s.t. } C_j(x) \geq 0, j = 1, \dots, m \end{aligned} \quad (3.8)$$

where $F(x)$ is the objective function and $C_j(x)$ is the constraint function. Both of them can be linear or nonlinear. One should notice that only inequality constraints are accepted and the initial point x_0 cannot be set to a zero vector. The solver used in this paper is from an open-source C# library, named CsNumeric ("CsNumeric," 2014).

3.5. Adaptive prediction horizon

A larger prediction horizon leads to moderate, but less accurate control actions. However, a smaller prediction horizon typically yields higher accuracy but aggressive control actions, especially when the difference between prediction and reference is significant. This can result in large overshoots and oscillations of the real track. To address this issue, the approach of an adaptive prediction horizon is proposed, which is quite simple but useful and is formulated as follows:

$$\begin{aligned} \text{if } T_e > 0.5B \text{ or } H_e > 5 \text{ deg}, N_p = K_{NP}N_{p0} \\ \text{else } N_p = N_{p0} \end{aligned} \quad (3.9)$$

where T_e is the track error and H_e is the heading error, N_{p0} is the default prediction horizon, and K_{NP} is a tuneable parameter larger than or equal to 1. The main strategy is to increase the prediction horizon when heading error and track error are large. When $K_{NP} = 1$, the prediction horizon is not adaptive.

3.6. Optimization efficiency

The computational efficiency of the two MPC optimizers, programmed in C#, were both tested on a laptop equipped with an Intel Core i7-10850H CPU of 2.70 GHz and 16 GB RAM. The performance of the optimizers was evaluated in terms of calculation time per time step, as presented in Table 6. The results indicate that the two optimizers exhibit almost equivalent execution speed. Additionally, compared with other optimizers reported in Table 4, both optimizers pose outstanding efficiency. This allows the resulting model predictive controllers to work in systems with high sampling frequencies.

4. Experimental free running model tests

4.1. Experimental setup

In order to validate the effectiveness of the proposed path following

Table 6
Calculation time per time step of the Cobyta and the Spline optimizers.

Optimizer	MPC type	Average calculation time [s]	Max calculation time [s]	Sampling time [s]	Prediction horizon/step
Cobyta	NMPC	0.0030	0.032	0.125	40
Spline	NMPC	0.0025	0.026	0.125	40

Table 7

Main dimensions of the towing tank.

Parameter	Total length [m]	Net length [m]	Width [m]	Maximum water depth [m]	Length of the ship models [m]
Value	174.0	140.0	20.0	1.0	3.5–8.0

algorithm, a series of free running model tests has been carried out in the Towing Tank for Manoeuvres in Shallow Water (Delefortrie et al., 2019) in Ostend, Belgium (Flanders Hydraulics, in co-operation with Ghent University). The dimensions of the towing tank are listed in Table 7. The width of the new tank (20 m) is almost three times the width (7 m) of the Towing Tank for Manoeuvres in Confined Water in Antwerp, Belgium, for the purpose of sufficiently eliminating tank wall effects. This allows a maximum model length of about 8 m. Fig. 3 shows the layout of the towing tank and also its coordinate system.

A free running system was deployed in this tank, which is a totally independent system for validating automatic control schemes. It is a ROS-based platform (robot operating system) ("ROS2," 2022), allowing conduction of a series of autonomous manoeuvres in the towing tank, such as acceleration, zigzag manoeuvres, heading/course keeping, path following and deceleration. The system comprises two main components, namely, the ship and the wall PC, which communicate with each other via WiFi. The ship performs free running tasks, while the wall PC functions as the remote control centre that allows users to issue commands in consoles and monitor ship states through a user-friendly interface. The system operates currently at a sampling frequency of 8 Hz. A schematic of the system layout is depicted in Fig. 4.

The KVLCC2 ship is a well-known tanker, which is widely used as a benchmark ship. A 1/75 scale model ("SIMMAN2020," 2023), as shown in Fig. 5, is employed to execute the free running experiments. Its position is measured by a lidar and an IMU (Inertial Measurement Unit). This ship only has one rudder and one propeller, making it an under-actuated ship, a common condition for surface ships. Table 8 lists the main principles of this ship model at tested draft. The tested water depth is 1 m, providing a deep water condition according to PIANC (PIANC, 1992).

About 70 free running tests have been carried out. A selection of the model tests are discussed here. In each test, a combination of three autonomous manoeuvres, accelerating – path following– decelerating, was involved. For the first and second manoeuvres, a constant ahead propeller rate, e.g. 300 RPM which was the default value in this paper, was set. Three routes are predefined to test the MPC path following algorithm as listed in Table 9.

4.2. Evaluation metrics for control performance

Chen (2021) performed a series of free running tests in the smaller Towing Tank for Manoeuvres in Confined Water to investigate the effectiveness of the proposed adaptive PID controller and the Fuzzy controller. Five indices were used to evaluate their performance. The research shown in this paper extends his work towards curved path following and a modern control method. The evaluation metrics were retained for quantitative assessment and comparison:

1. Mean track error integral, mTEI (tracking accuracy)

$$\text{mTEI} = \frac{1}{N_1 - N_0 + 1} \sum_{k=N_0}^{N_1} |T_e(k)|$$

2. Maximum track error, MTE (tracking accuracy)

$$\text{MTE} = \max\{|T_e(N_0)|, |T_e(N_0 + 1)|, \dots, |T_e(N_1)|\}$$

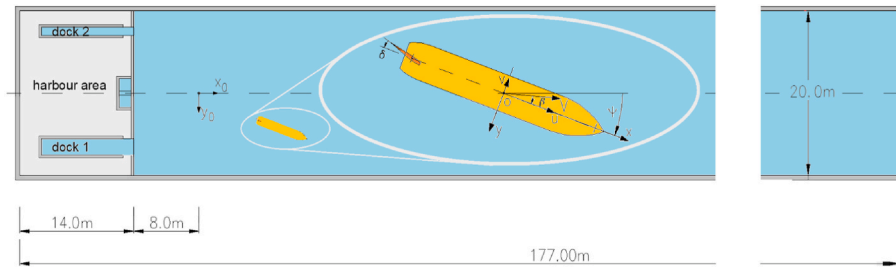


Fig. 3. Towing tank main dimensions and its coordinate systems.

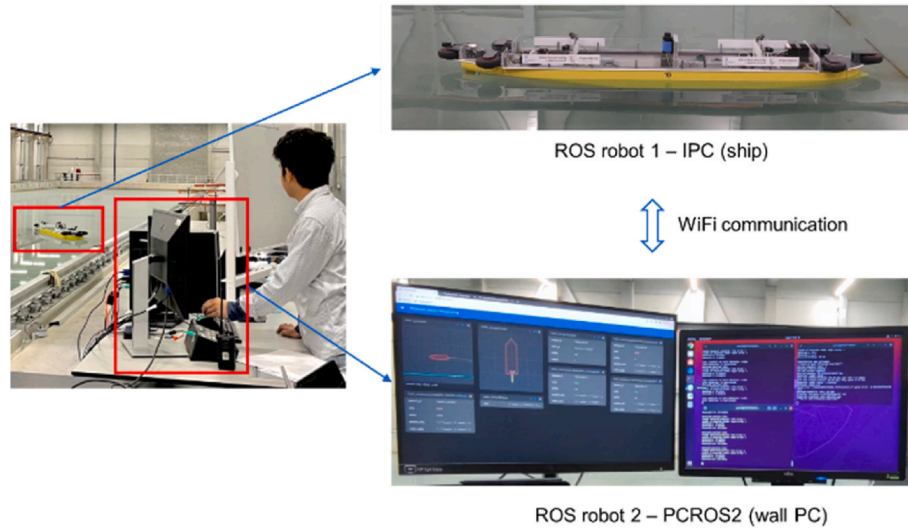


Fig. 4. Layout of the free running system.

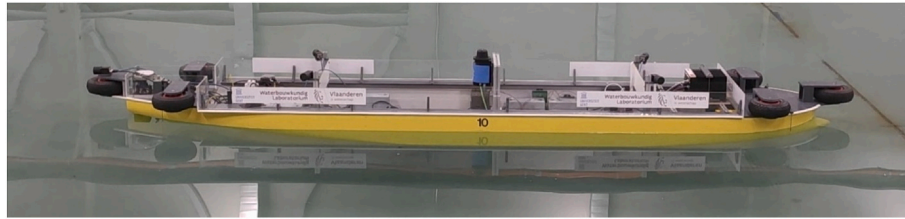


Fig. 5. 1/75 scale model of KVLCC2 benchmark ship.

3. Mean rudder integral, mRI (aggressiveness)

$$\text{mRI} = \frac{1}{N_1 - N_0 + 1} \sum_{k=N_0}^{N_1} |\delta(k)|$$

4. Mean rudder total variation, mRTV (energy consumption)

$$\text{mRTV} = \frac{1}{N_1 - N_0} \sum_{k=N_0+1}^{N_1} |\delta(k) - \delta(k-1)|$$

Table 8
Main dimensions of KVLCC2 at tested draft.

Parameter	Value	Parameter	Value
Length between perpendiculars, L_{pp} [m]	4.267	Breadth at design water line, B_{wl} [m]	0.773
Depth of a ship hull, D [m]	0.4	Draft, T [m]	0.277
Displacement [m^3]	0.741	Propeller rotation	Right handed
Maximum RPM	1000	Maximum rudder angle [deg]	35
Turn rate of rudder [deg/s]	20.9	L_{CG} [m]	2.281

Table 9
Definition of predefined paths.

Name	Path type	Waypoints	Path direction
Path A	Simple	(20, 4), (30, 4), (50, 0), (70, 0), (90, 6), (120, 6)	From harbour
Path B	Simple	(120, 6), (90, 6), (70, 0), (50, 0), (30, 4), (20, 4)	Towards harbour
Path C	Complex	(20, 4), (40, 4), (60, 6), (80, 0), (100, -8), (110, 0), (100, 8), (80, 0), (60, -6), (30, 0), (8, 8), (1, 4), (5, 0)	From harbour and then back to harbour

5. Mean heading error integral

$$\text{mHEI} = \frac{1}{N_1 - N_0 + 1} \sum_{k=N_0}^{N_1} |H_e(k)|$$

In the course of conducting free running tests, the installation work of the carriage rendered the harbour inaccessible for most of the time, and consequently the free-sailing ship was usually initialized outside the harbour with an uncertain attitude. Then the initial position and heading error were not exactly controllable which adds difficulty for the fair evaluation. However, the motion control system implemented in the ship ensures that it gradually reduces the initial heading and position error over time. As a result, by the time the ship reaches the third waypoint, it has typically achieved a similar attitude as desired. Therefore, the control performance and the route complexity were evaluated with only a segment of the path that extended from the third waypoint to the end. In addition, for the complex path, Path C, as the last two waypoints were just defined for turning the ship to a suitable attitude for the subsequent run, this segment of path was also excluded from the evaluation.

4.3. Indexes for describing path complexity

In addition, two indices were adopted to describe the complexity of the predefined routes:

1. Mean curvature integral, mCI

$$\text{mCI} = \frac{1}{N_1 - N_0 + 1} \sum_{k=N_0}^{N_1} \kappa(k)$$

which calculates the mean curvature of the curved path. Larger curvature means more direction vibration and larger difficulty to follow the path.

2. Maximum curvature, MC

$$\text{MC} = \max\{\kappa(N_0), \kappa(N_0 + 1), \dots, \kappa(N_1)\}$$

which represents the biggest difficulty to follow on the path.

Table 10 lists the evaluated complexity of the predefined paths mentioned in Section 4.1. It is clear that the ranking of the path complexity is: Path A < Path B < Path C.

4.4. Parameter setting

In path following, the control of velocities is not required, thus the weight matrix, $\mathbf{Q}$, is defined as:

$$\mathbf{Q} = \text{diag}[1, 1, W_H, 0, 0, 0]$$

where W_H represents the weight of the heading error. The position error weights for x and y are set to the same value since both errors are equally important. They are set to 1 to fix these weights, leaving only the weight of heading error to be tuned.

Table 10

Evaluation of complexity of predefined paths (WPT_{*i*} denotes the *i*-th waypoint and *M* is the number of waypoints).

Name	Path type	Path direction	Evaluated part	mCI	MCI
Path A	Simple	Harbour – wave maker	WPT ₃ ~ WPT _{<i>M</i>}	0.0130	0.0308
Path B	Simple	Wave maker – harbour	WPT ₃ ~ WPT _{<i>M</i>}	0.0165	0.0453
Path C	Complex	Harbour – wave maker – harbour	WPT ₃ ~ WPT _{<i>M-2</i>}	0.0411	0.1663

The performance of MPC is closely related to its prediction. On one hand, an accurate predictive model is preferable. Its model coefficients needs to be determined for each ship, i.e. K , T and B for the Nonlinear Nomoto model introduced in Section 3.3. These parameters were identified by using linear least squares method driven by data collected from zigzag tests, e.g. a 20/20 zigzag. In some cases, the B value was also estimated to account for the effect of asymmetrical hulls or asymmetry induced by propulsion. As the KVLCC2 ship turns, with a zero rudder angle, to her starboard side due to the so called wheel effect, a negative B should be set. With a constant rudder angle, $\dot{r}$ tends to zero at the stable stage and the yaw rate will approximately equal to $K\delta - B$. From observation during the acceleration test, a rudder angle of about two degrees needs to be performed to compensate the asymmetrical effects of the propeller. Therefore, it can be estimated by

$$B \approx \frac{2\pi}{180} K$$

For instance, B is estimated to -0.0022 with $K = -0.064$, $T = 16$. On the other hand, the prediction horizon, N_p , also affects the quality of the prediction, as described in Section 3.2. From experience, its value could be set preliminarily to $1/K$, T or L_{pp}/U and be fine-tuned based on the control performance, e.g. decreasing it by 25% if mTEI is high, or increasing it by 25% if mTEI is low but mRI and mRTV are high. Table 11 lists the essential parameters in MPC and explains the effects of them, which will be validated and discussed in detail in the next chapter.

The guidance system includes an additional parameter, K_1 , which controls the convergence rate of the along-track error with respect to SF frame and affects the reference generation. A larger K_1 results in faster convergence. During experiments, a significant reduction in tracking error was observed when K_1 was set from 0.01 to 0.1. Consequently, a value of 0.1 was preferred for K_1 . As the SF origin is regarded as a particle and its moving speed can easily converge to the desired one, this parameter is not required to be fine-tuned, nonetheless it should be assigned to a large enough value to ensure the SF frame can quickly follow the ship. Its value can be fixed for similar ships.

5. Experimental results

5.1. Weight factor for heading error

W_H is set to several values ranging from 0 to 1, while other MPC parameters are held constant for the sake of comparison to investigate the impact of varying the weight factor for the heading error (W_H) on the performance of the MPC path-following controller. Two predefined paths, Path A and Path B, are employed for path following, and some

Table 11

Essential parameters in MPC.

Parameter	Description	Influence	Value
W_H	Weight factor for heading error	Larger value increases the accuracy of tracking the reference heading, but reduces that of tracking the reference position	0–1
K, T, B	Parameters of predictive model	Determine prediction accuracy	Identified/estimated
N_p	Prediction horizon	Smaller value leads more aggressive control actions, less calculation time and possibly higher prediction accuracy	Around $1/K$, T or L_{pp}/U
CRT	Tolerance of cost reduction (Spline optimizer, optional)	Decrease the frequency of rudder deflection, but also decrease the control accuracy in some extent	0–5%
K_{NP}	Gain of adaptive prediction horizon	Increase the prediction horizon when heading error and track error are large	≥ 1

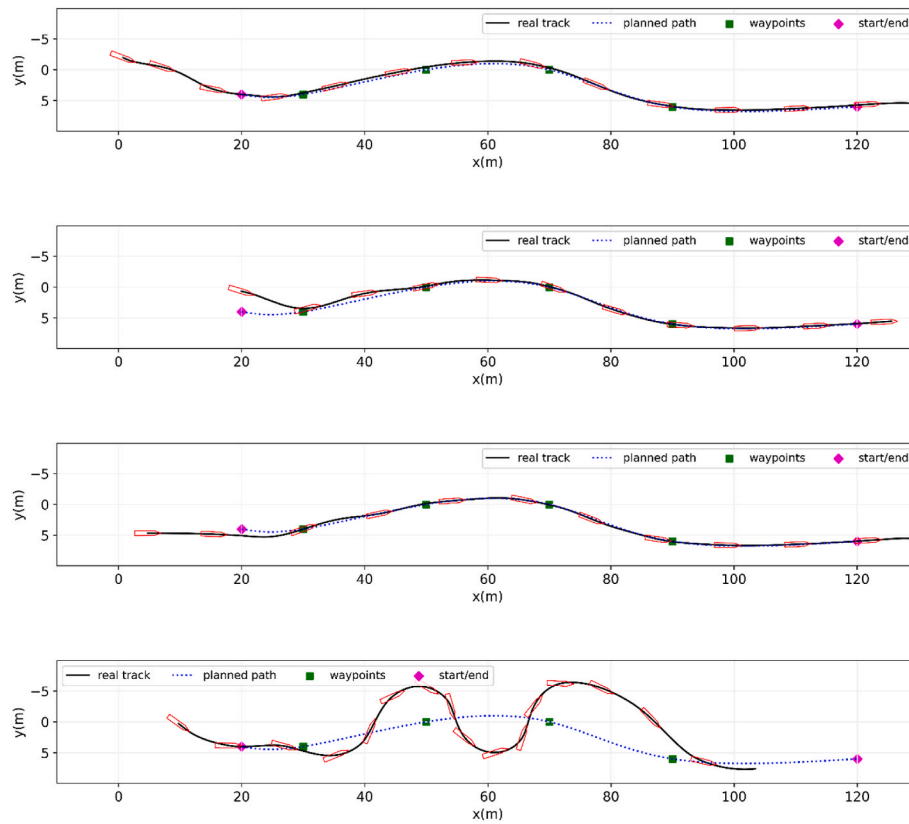


Fig. 6. Following Path A with the settings: $W_H = 1, 0.2, 0.1$ or 0 (top to bottom: No. 2022-11-07-06, 2022-09-29-01, 2022-10-05-07 and 2022-11-07-08), $K = -0.064$, $T = 16$, $B = -0.0022$, $N_{p0} = 10$, optimizer = Colyba, RPM = 300.

Table 12

MPC control performance with different settings of the weight of heading error ($K = -0.064$, $T = 16$, $B = -0.0022$, $N_{p0} = 10$, optimizer = Colyba).

Test No.	W_H	mTEI [m]	MTE [m]	mRI [deg]	mRTV [deg/s]	mHEI [deg]	Planned path
2022-11-07-06	1	0.2259	0.4286	3.5128	1.2567	0.7269	Path A
2022-09-29-01	0.2	0.1097	0.2817	5.8441	2.9921	0.8821	Path A
2022-10-05-07	0.1	0.0687	0.1919	5.7488	1.8314	0.8388	Path A
2022-11-07-08	0	4.0593	7.7623	22.6874	1.0447	37.4572	Path A
2022-09-26-11	1	0.3080	1.1947	7.8704	1.6957	2.2456	Path B
2022-11-07-07	0.2	0.1961	0.3205	7.5801	1.8424	1.1705	Path B
2022-10-05-08	0.1	0.1123	0.2737	8.2372	2.1035	1.1961	Path B

resultant tracks as measured in the towing tank are illustrated in Fig. 6. Evaluation metrics are provided in Table 12, and the results demonstrate that decreasing W_H from 1 to 0.1 leads to a reduction in mean tracking error. Notably, all tests except for the test with $W_H = 0$ exhibit errors that are less than half of the ship's breadth. However, it is recommended to avoid setting W_H to 0, as this may result in aggressive operation and poor tracking performance. A lower W_H value could be chosen to improve tracking accuracy, but it is worth noting that decreasing W_H may come at a cost of increased energy consumption (larger mRTV) and large rudder response (larger mRI). The observations can be explained: the position errors are reduced by deflecting the heading. When W_H is set to a large number, minimizing heading errors is emphasized and the ship is asked more to keep courses parallel to the tangents of the path (reference headings). Therefore the position errors cannot be well eliminated. However if W_H is too small, with emphasis placed on minimizing position error, the headings will be deflected largely and frequently to minimize the position errors, resulting in huge oscillation as observed in the bottom subplot in Fig. 6. A trade-off needs to be played with setting this parameter.

5.2. Prediction horizon

Initially, a value of 15 s (close to T) for N_p was considered based on Table 10. However, the ship's tracking accuracy was not quite satisfactory when following Path A, as evidenced in the top subplot of Fig. 7. Subsequently, N_p was decreased to 10, 8, and 5 s, respectively, to assess the potential for improvement. In addition, the ship was required to follow Path B to examine the controller's performance when the planned path was reversed. Some results of these free running tests are compared in Fig. 7, and the corresponding evaluation metrics are summarized in Table 13. The statistics indicate that all mTEI values were smaller than half the ship's breadth, with the lowest value being achieved at $N_p = 8$ s. It is noteworthy that a trend of improving tracking accuracy is observed with a smaller prediction horizon. However, a very small prediction horizon could result in aggressive manoeuvring behaviour, i.e. larger and more frequent control actions, leading to large overshoots and oscillations, and reduced tracking accuracy. For example, when $N_p = 5$, mRI and mRTV were significantly large, and mTEI and MTE were notably increased. Therefore, it is recommended to select a value between the max over which unacceptable tracking accuracy is delivered and the min below which obvious oscillations can be observed, and

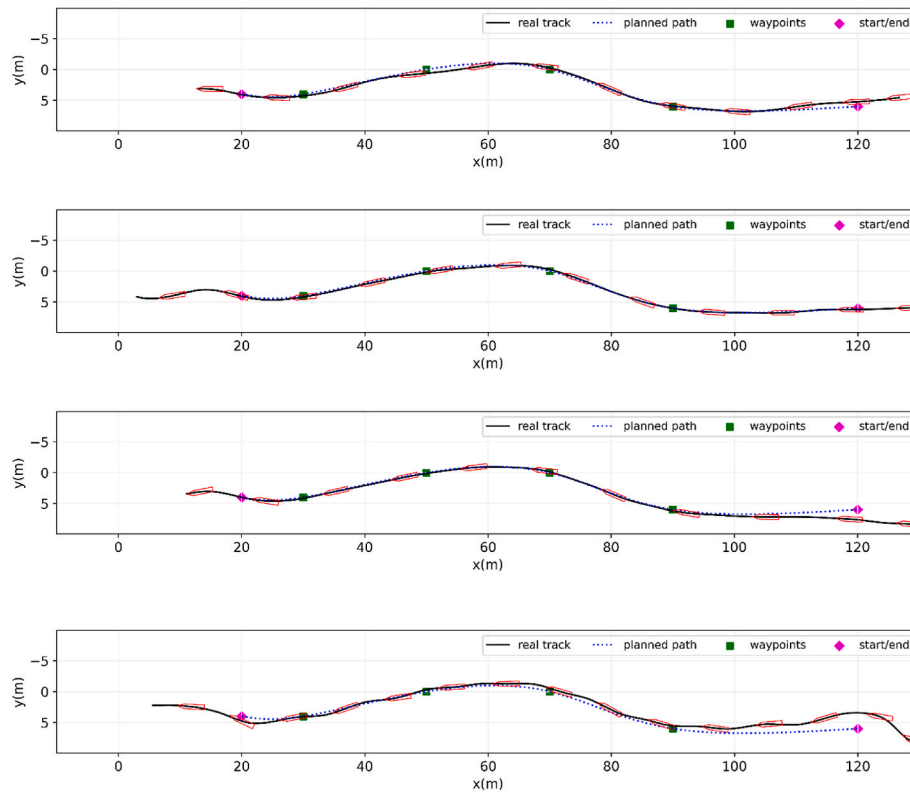


Fig. 7. Following Path B with the settings: $N_{p0} = 15, 10, 8$ or 5 (top to bottom: No. 2022-11-07-09, 2022-11-07-07, 2022-09-29-08 and 2022-11-07-11), $K = -0.064$, $T = 16$, $B = -0.0022$, $W_H = 0.2$, optimizer = Colyba, RPM = 300.

Table 13

MPC control performance with different settings of prediction horizon ($K = 0.064$, $T = 16$, $B = -0.0022$, $W_H = 0.2$, optimizer = Colyba).

Test No.	N_{p0}	mTEI [m]	MTE [m]	mRI [deg]	mRTV [deg/s]	mHEI [deg]	Planned path
2022-09-29-03	15	0.1728	0.4077	5.4632	1.3532	1.8467	Path A
2022-09-29-01	10	0.1097	0.2817	5.8441	2.9921	0.8821	Path A
2022-11-07-10	8	0.0871	0.1911	5.9435	2.5736	0.6894	Path A
2022-11-07-12	5	0.2230	0.5846	30.3797	3.6187	5.6575	Path A
2022-11-07-09	15	0.2897	0.7402	7.3075	1.3933	2.6293	Path B
2022-11-07-07	10	0.1961	0.3205	7.5801	1.8424	1.1705	Path B
2022-09-29-08	8	0.1061	0.1830	6.7075	2.5072	0.7814	Path B
2022-11-07-11	5	0.2981	0.8464	29.9862	4.0953	4.3042	Path B

make a trade-off between the tracking accuracy and moderate control behaviour.

5.3. Adaptive prediction horizon

The primary objective of this section is to validate the effectiveness of utilizing an adaptive prediction horizon by focusing on observing the smoothness of the ship's transition to the planned path when there are significant heading and position errors. As depicted in Fig. 8, the adoption of an adaptive prediction horizon ($K_{NP} = 1.5$ or 2) led to smoother processing of large heading and position errors, which was evidenced by a reduction in overshoot (in the area between the first and second waypoints, the position errors and heading errors were both reduced as K_{NP} increased). This outcome can be attributed to the ability of the controller to look ahead to a longer path section when the ship encounters significant deviations from the planned path and the controller is equipped with the adaptive prediction horizon, thereby allowing it to identify and implement optimal control actions that result in minimal overshoot.

5.4. Predictive model

Different sets of predictive model parameters were used to investigate their effects. By setting zero and non-zero values to the parameter B in Eq. (3.6), the difference between implementing a conventional Nomoto model or a nonlinear Nomoto model can be compared. The nonlinear Nomoto model was expected to perform better for ships with asymmetric behaviour.

Table 14 shows that there was a significant difference in T parameter setting (12.6919 and 16) among the various sets of predictive model parameters examined. These values were identified with two different zigzags, 20/20 zigzag and 20/5 zigzag, respectively, and the identified models could predict the yaw rate well as seen in Fig. 9. However, all four settings of Nomoto parameters delivered high tracking accuracy, with the test No. 2022-11-07-10 exhibiting the best tracking accuracy. This means the tracking accuracy is insensitive to the T parameter. The real tracks can be seen in Fig. 10. Even though, one still needs to notice that rudder behaviour was more aggressive and deflected frequently in the test No. 2022-11-07-20, indicating that good tracking accuracy was maintained through paying huge effort on manoeuvring the rudder sometimes when an preferable setting was not given.

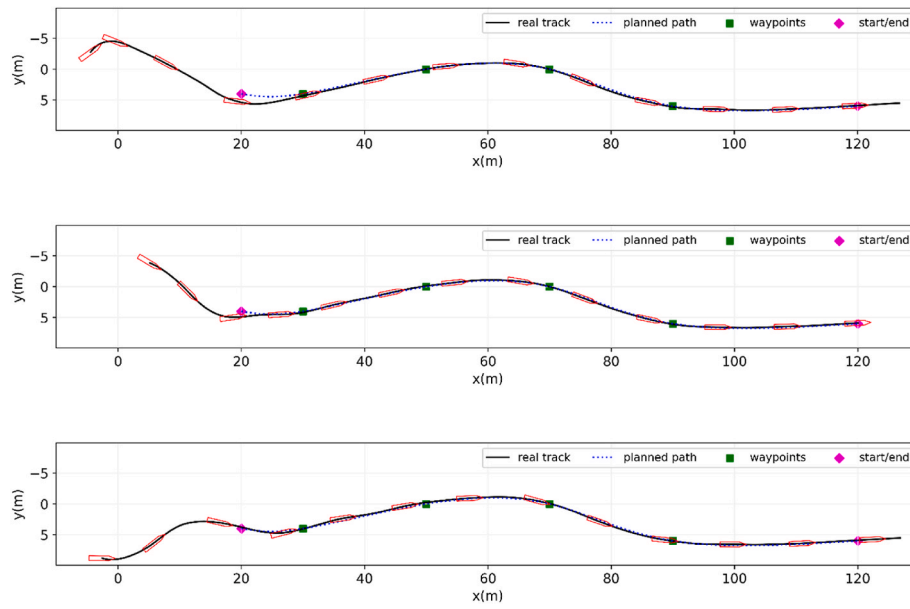
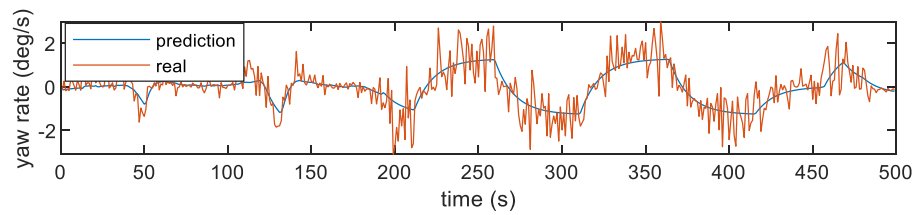


Fig. 8. Following Path A with the settings: $K_{NP} = 1, 1.5$ or 2 (top to bottom: No. 2022-11-07-14, 2022-11-14-13 and 2022-11-07-16), $N_{p0} = 10$, $K = -0.064$, $T = 16$, $B = -0.0022$, $W_H = 0.2$, optimizer = Colyba, RPM = 300.

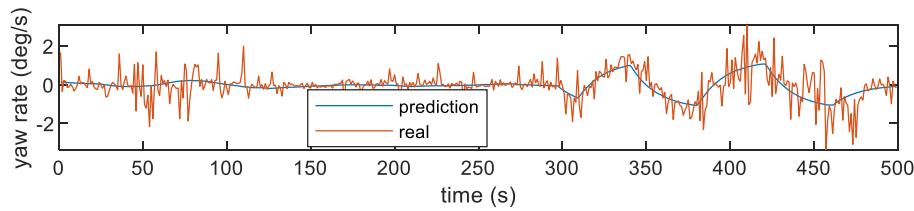
Table 14

MPC control performance with different settings of Nomoto parameters ($N_p = 8$, $W_H = 0.2$, optimizer = 1VarColyba).

Test No.	Nomoto parameters	mTEI [m]	MTE [m]	mRI [deg]	mRTV [deg/s]	mHEI [deg]	Planned path
2022-09-29-07	$K = -0.0665$ $T = 12.6919$ $B = -0.0010$	0.1007	0.2902	5.8540	1.9831	0.8471	Path A
2022-11-14-15	$K = -0.0665$ $T = 12.6919$ $B = 0$	0.1344	0.4067	7.5268	2.2646	1.1018	Path A
2022-11-07-10	$K = -0.064$ $T = 16$ $B = -0.0022$	0.0871	0.1911	5.9435	2.5736	0.6894	Path A
2022-11-07-20	$K = -0.064$ $T = 16$ $B = 0$	0.0959	0.2687	9.0623	7.0163	0.8317	Path A



(a) 20/20 zigzag, Nomoto model: $K = -0.0665$, $T = 12.6919$, $B = 0$



(b) 20/5 zigzag, Nomoto model: $K = -0.064$, $T = 16$, $B = 0$

Fig. 9. Comparison between prediction and measured data.

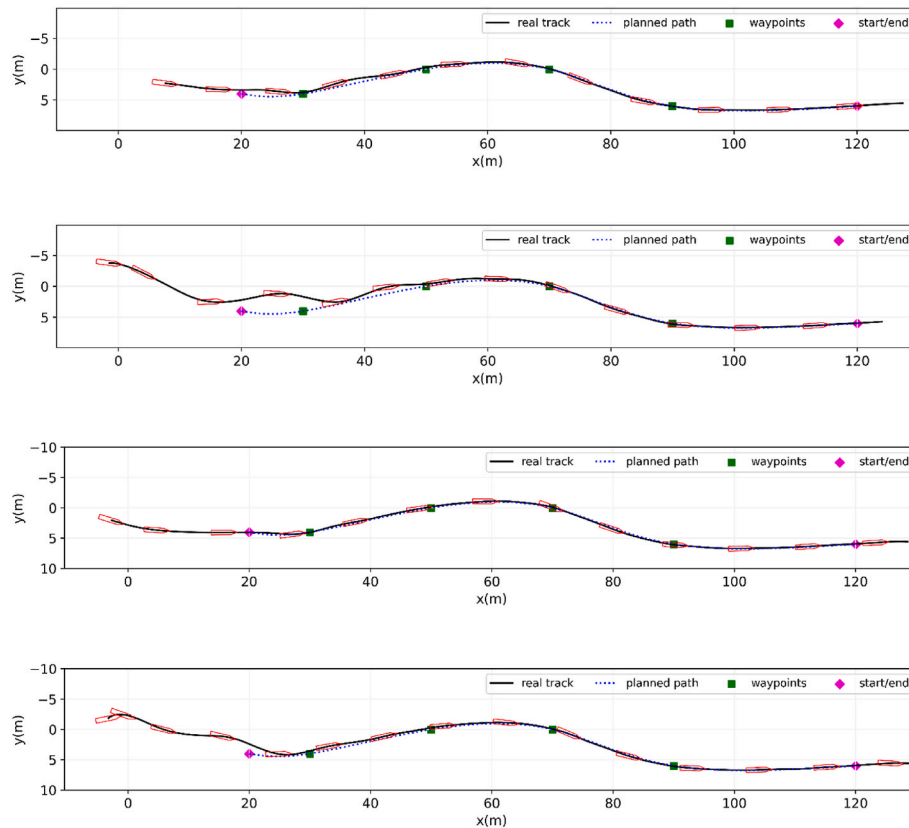


Fig. 10. Trajectories when following Path A with the settings: Nomoto model = $\{K = -0.064, T = 16, B = -0.0022\}$, $\{K = -0.0665, T = 12.6919, B = 0\}$, $\{K = -0.064, T = 16, B = -0.0022\}$ or $\{K = -0.064, T = 16, B = 0\}$ (top to bottom: No. 2022-09-29-07, 2022-11-14-15, 2022-11-07-10, 2022-11-07-20), $N_{p0} = 8$, $W_H = 0.2$, optimizer = Colyba, $K_{NP} = 1.5$, RPM = 300.

The nonlinear (modified) model outperformed the linear model by slightly increasing tracking accuracy (smaller mTEI and MTE) but significantly improving rudder performance, i.e. presenting moderate rudder behaviour (smaller mRI) and lower energy consumption (smaller mRTV). The rudder performance can be visually compared in Fig. 11.

5.5. Optimizer

Two optimizers, the Spline optimizer and the Coby optimizer, are introduced in Section 3.4. Their respective performance is compared in this section, as listed in Table 15. From the table, it is not really clear that which optimizer performed better. Both optimizers could deliver good control performance. When $CRT = 0$, the optimizer Spline's performance was quite close to Coby (similar tracking accuracy and rudder behaviour). However, with a non-zero CRT , differences in rudder performance emerged between the two optimizers. The Spline optimizer exhibited more aggressive rudder behaviour with a larger mRI. Although the mRTV remained similar to that of the Coby optimizer, Fig. 13 clearly illustrates that the Spline optimizer resulted in less frequent rudder performance. Fig. 12 demonstrates the real tracks when implement different optimizers to follow Path B.

5.6. Planned path

The MPC controller was applied to a more complicated scenario of following Path C. This predefined path required the ship to take a round trip in the towing tank. The most difficult part was the turning of 180° to reverse the navigation direction. The experimental results, depicted in Fig. 14, demonstrates the excellent performance of the MPC, as the ship navigated the predefined big turning part with high accuracy. This test is evaluated in Table 16, which also lists the metrics for following the

simple paths, Path A and B. As expected, the tracking accuracy is relatively lower for following a more complicated path. This proves that the MPC path follower can be applied to complex environments to steer the ship to very complex paths with high accuracy.

5.7. Robustness

The Nomoto model is designed to function at a specific speed and is incapable of adapting to varying RPMs. As a consequence, a significant change in RPM results in a considerable deviation between the prediction of MPC and actual system responses. Nonetheless, an effective control design should demonstrate robustness to internal model uncertainties arising from the inherent limitations of the Nomoto model. This section explores how the MPC controller utilizing a fixed predictive model acts to significant change in rotational speed. Specifically, three rotational speeds – 200 rpm, 300 rpm, and 400 rpm – were adopted as operating conditions, with 300 rpm considered the normal operating situation, as all control parameters were determined under this condition.

Table 17 presents the controller's performance under these conditions when implementing same MPC parameters and the trajectories are visualized in Fig. 15. The statistics reveal that all mTEIs are less than a third of the ship's breadth, indicating high accuracy can be maintained despite changes in operating propeller rates (increasing or decreasing by 100 rpm). This highlights the strong robustness of the MPC controller. Increasing the propeller rate even improves tracking accuracy. However, decreasing the rate adversely affects both tracking accuracy and rudder behaviour, but the deterioration remains within acceptable limits. This phenomenon is reasonable given that a ship's manoeuvrability typically decreases at lower speeds as lower propeller rates resulting in lower lift forces on the rudder. More evaluated tests are

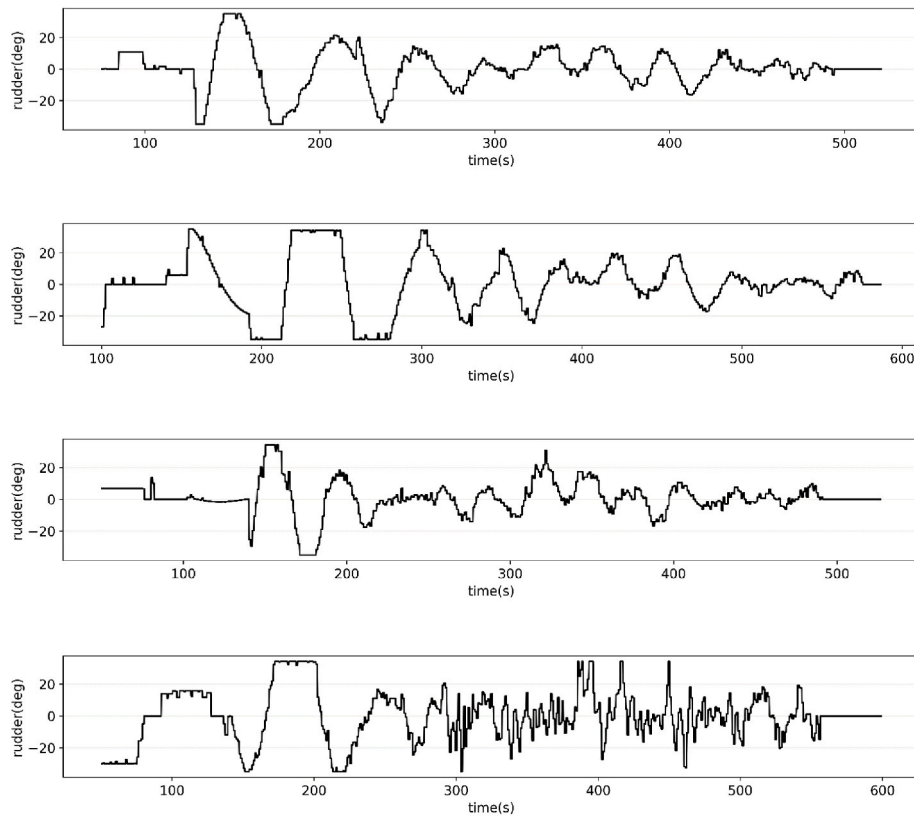


Fig. 11. Time histories of measured rudder angle when following Path A with the settings: Nomoto model = $\{K = -0.064, T = 16, B = -0.0022\}$, $\{K = -0.0665, T = 12.6919, B = 0\}$, $\{K = -0.064, T = 16, B = -0.0022\}$ or $\{K = -0.064, T = 16, B = 0\}$ (top to bottom: No. 2022-09-29-07, 2022-11-14-15, 2022-11-07-10, 2022-11-07-20), $N_{p0} = 8$, $W_H = 0.2$, optimizer = Colyba, $K_{NP} = 1.5$, RPM = 300.

Table 15

MPC control performance with different optimizers ($K = 0.064$, $T = 16$, $B = -0.0022$, $N_p = 10$, $W_H = 0.2$).

Test No.	Optimizer	CRT	mTEI [m]	MTE [m]	mRI [deg]	mRTV [deg/s]	mHEI [deg]	Planned path
2022-09-29-01	1VarColyba	–	0.1097	0.2817	5.8441	2.9921	0.8821	Path A
2022-11-07-21	1VarSpline	0.05	0.0926	0.2713	7.4714	2.8249	1.0937	Path A
2022-10-05-01	1VarSpline	0.01	0.0813	0.1678	6.0528	2.2434	0.9933	Path A
2022-11-07-22	1VarSpline	0	0.1169	0.3408	6.0780	2.5954	0.9967	Path A
2022-11-07-07	1VarColyba	–	0.1961	0.3205	7.5801	1.8424	1.1705	Path B
2022-09-29-10	1VarSpline	0.05	0.1623	0.3517	13.7316	2.2359	2.1291	Path B
2022-11-07-19	1VarSpline	0.01	0.1872	0.3275	9.5832	1.8872	1.5300	Path B
2022-10-05-02	1VarSpline	0	0.1422	0.2235	7.4153	2.2617	1.0088	Path B

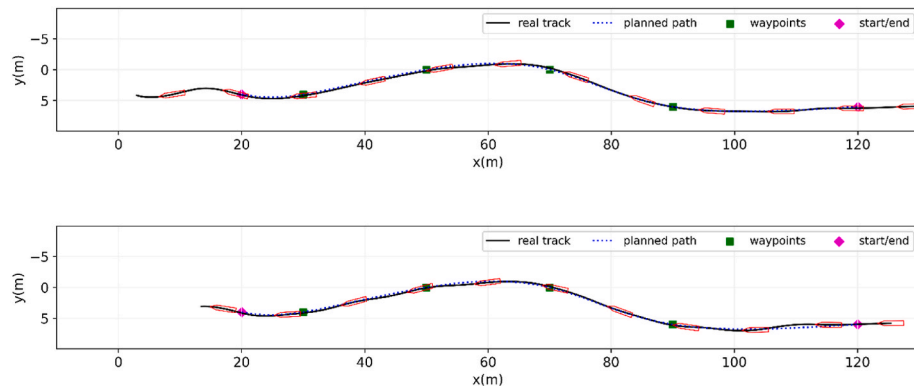


Fig. 12. Trajectories when following Path B with the settings: optimizer = Colyba or optimizer = Spline with $CRT = 0.05$ (top to bottom: No. 2022-11-07-07, 2022-09-29-10), $K = -0.064$, $T = 16$, $B = -0.0022$, $N_{p0} = 10$, $K_{NP} = 1.5$, $W_H = 0.2$, RPM = 300.

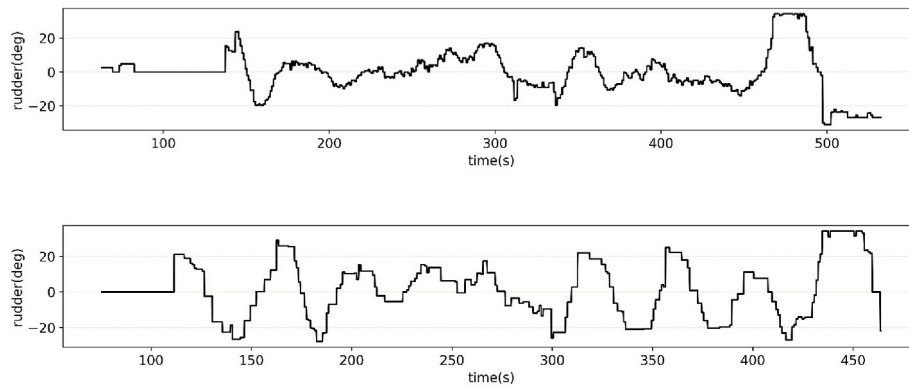


Fig. 13. Time histories of the measured rudder angle when following Path B with the settings: optimizer = Colyba or optimizer = Spline with $CRT = 0.05$ (top to bottom: No. 2022-11-07-07, 2022-09-29-10), $K = -0.064$, $T = 16$, $B = -0.0022$, $N_{p0} = 10$, $K_{NP} = 1.5$, $W_H = 0.2$, $RPM = 300$.

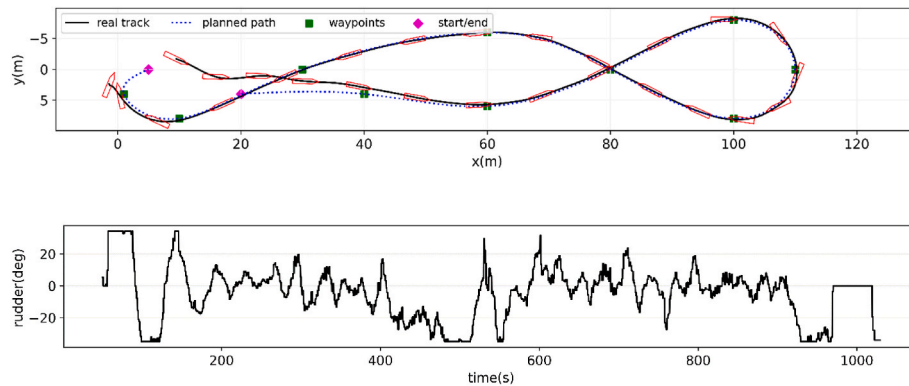


Fig. 14. Trajectory and time history of measure rudder angle when following Path C with the setting (No. 2022-10-05-12): optimizer = Colyba, $K = -0.064$, $T = 16$, $B = -0.0022$, $N_{p0} = 8$, $K_{NP} = 1.5$, $W_H = 0.2$, $RPM = 300$.

shown in Table 18.

6. Conclusion

This paper presents a practical MPC path following controller for autonomous ships, which employs a course model with very few parameters for the predictive model and poses two different optimizers to retrieve real-time optimal control actions efficiently. Experimental tests conducted in the towing tank validate the effectiveness of the controller and demonstrate its robustness within a specific range of speeds. The results prove the controller has the capability to accurately steer a ship along both simple and complex reference paths while carrying out appropriate control behaviours. Involved parameters are explained in detail and comparative studies conducted to investigate the impact of

different parameter settings reveal the following findings:

1. A large prediction horizon or weight factor assigned to the heading error reduces the tracking accuracy, but extremely low values result in aggressive and power intensive control actions, exhibiting large oscillation in the performed trajectory. Therefore, a trade-off between tracking accuracy and moderate control behaviours should be considered.
2. The adoption of an adaptive prediction horizon strategy allows for smoother handling of large heading and position errors.
3. The use of a modified course model that accounts for ship's asymmetric behaviour yields better performance.

In the near future, the robustness of the proposed MPC path follower

Table 16

MPC control performance with respect to different planned paths ($K = 0.064$, $T = 16$, $B = -0.0022$, $N_p = 8$, $W_H = 0.2$).

Test No.	Path type	mTEI [m]	MTE [m]	mRI [deg]	mRTV [deg/s]	mHEI [deg]	Planned path	mCI	MCI
2022-11-07-10	simple	0.0871	0.1911	5.9435	2.5736	0.6894	Path A	0.0130	0.0308
2022-09-29-08	simple	0.1061	0.1830	6.7075	2.5072	0.7814	Path B	0.0165	0.0453
2022-10-05-12	complex	0.1733	0.4631	10.8061	2.6780	1.0363	Path C	0.0411	0.1663

Table 17

MPC control performance in different RPM.

Test No.	RPM	N_p	W_H	mTEI [m]	MTE [m]	mRI [deg]	mRTV [deg/s]	mHEI [deg]
2022-11-07-05	200	10	0.1	0.1963	0.6822	16.6510	2.7609	2.4140
2022-10-05-12	300	10	0.1	0.1883	0.4803	9.1523	1.9331	1.2933
2022-11-07-01	400	10	0.1	0.1056	0.3099	11.6151	2.6633	1.4039

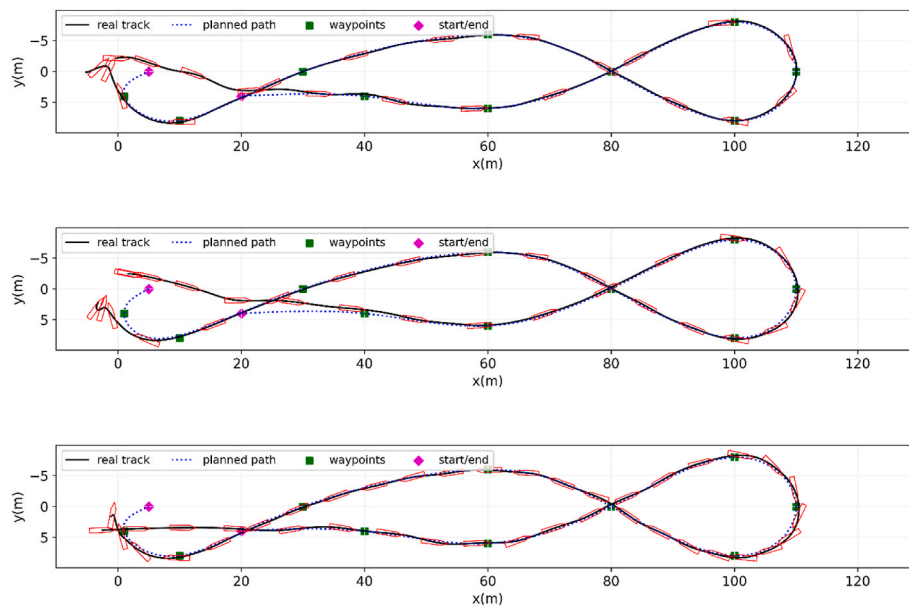


Fig. 15. Trajectories when following Path C with the settings: RPM = 400, 300 or 200 (No. 2022-11-07-05, 2022-10-05-12, 2022-11-07-01), optimizer = Colyba, $K = -0.064$, $T = 16$, $B = -0.0022$, $N_{p0} = 10$, $K_{NP} = 1.5$, $W_H = 0.1$.

Table 18

MPC control performance in different RPM, N_p and W_H .

Test No.	RPM	N_p	W_H	mTEI [m]	MTE [m]	mRI [deg]	mRTV [deg/s]	mHEI [deg]
2022-11-14-17	200	12	0.1	0.2063	0.6159	12.3313	1.9934	2.2988
2022-11-07-02	300	10	0.2	0.1560	0.3136	10.9745	2.1701	1.2101
2022-11-07-03	300	8	0.2	0.1733	0.4631	10.8061	2.6780	1.0363
2022-11-07-04	300	8	0.1	0.1121	0.3924	13.2609	3.7481	1.3006
2022-10-05-15	400	10	0.2	0.1448	0.3498	8.8671	1.4941	0.9512
2022-10-05-17	400	8	0.2	0.1341	0.5367	10.4370	2.3909	1.1778
2022-10-05-18	400	8	0.1	0.0931	0.4435	14.6429	3.3530	1.5560

under varying water depths and across a wider range of speeds will be investigated. Additionally, the development of an adaptive predictive model could further enhance the robustness of the MPC system. Furthermore, trajectory tracking and collision avoidance will be co-designed with the MPC path follower to expand its applicability in different scenarios. These tests and enhancements will provide valuable insights into the performance and versatility of the MPC-based control system.

CRedit authorship contribution statement

Hongwei He: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Thibaut Van Zwijnsvoorde:** Conceptualization, Formal analysis, Investigation, Resources, Data curation, Writing – review & editing, Visualization, Supervision, Project administration. **Evert Lataire:** Conceptualization, Resources, Data curation, Writing – review & editing, Supervision, Project administration. **Guillaume Delefortrie:** Conceptualization, Methodology, Software, Validation, Investigation, Resources, Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgement

This work is supported by the program of the China Scholarship Council (202106230111).

References

- Abdelaal, M., Fränzle, M., Hahn, A., 2018. Nonlinear model predictive control for trajectory tracking and collision avoidance of underactuated vessels with disturbances. *Ocean Eng.* 160, 168–180. <https://doi.org/10.1016/j.oceaneng.2018.04.026>.
- Akdag, M., Fossen, T., Johansen, T., 2022. Collaborative collision avoidance for autonomous ships using informed scenario-based model predictive control. *IFAC-PapersOnLine*. <https://doi.org/10.1016/j.ifacol.2022.10.439>.
- Chen, C., 2021. Numerical and Experimental Study on Ship Motion Control Systems in Shallow Water. Ghent University.
- CsNumeric [WWW Document]. URL: <https://github.com/cureos/csnumerics/wiki/COBYLA> (accessed 4.18.23).
- Delefortrie, G., Geerts, S., Lataire, E., Troch, P., Monbaliu, J., 2019. Coastal and ocean basin and towing tank for manoeuvres in shallow water at Flanders Maritime Laboratory. In: *The 6th International Conference on Advanced Model Measurement Technology for the Maritime Industry (AMT'19)*. Roma, Italy.
- Deng, F., Wang, L., Jiao, D., 2017. Adaptive observer based backstepping controller design for dynamic ship positioning. *China Ocean Eng.* 31, 639–645. <https://doi.org/10.1007/s13344-017-0073-7>.
- Fossen, T.I., 2011. *Handbook of Marine Craft Hydrodynamics and Motion Control*. John Wiley & Sons.
- Frenet, F., 1852. Sur les courbes à double courbure. *J. Math. Pure Appl.* 437–447.

- Ghaemi, R., Oh, S.-R., Sun, J., 2010. Path following of a model ship using Model Predictive Control with experimental verification. *Proc. 2010 Am. Control Conf.* <https://doi.org/10.1109/acc.2010.5530465>.
- Guerreiro, B.J., Silvestre, C., Cunha, R., Pascoal, A.M., 2014. Trajectory tracking nonlinear model predictive control for autonomous surface craft. *IEEE Trans. Control Syst. Technol.* <https://doi.org/10.1109/tcst.2014.2303805>.
- Hagen, I.B., Kufolalor, D., Johansen, T., Brekke, E., 2022. Scenario-based model predictive control with several steps for colregs compliant ship collision avoidance. *IFAC-PapersOnLine*. <https://doi.org/10.1016/j.ifacol.2022.10.447>.
- Johansen, T.A., Cristofaro, A., Perez, T., 2016. Ship collision avoidance using scenario-based model predictive control. *null*. <https://doi.org/10.1016/j.ifacol.2016.10.315>.
- Li, R., Huang, J., Pan, X., Hu, Q., Huang, Z., 2020a. Path following of underactuated surface ships based on model predictive control with neural network. *Int. J. Adv. Rob. Syst.* <https://doi.org/10.1177/1729881420945956>.
- Li, S., Liu, J., Negenborn, R.R., Negenborn, R.R., Wu, Q., 2020b. Automatic docking for underactuated ships based on multi-objective nonlinear model predictive control. *IEEE Access*. <https://doi.org/10.1109/access.2020.2984812>.
- Li, Z., Li, R., Bu, R., 2020c. Path following of under-actuated ships based on model predictive control with state observer. *J. Mar. Sci. Technol.* <https://doi.org/10.1007/s00773-020-00746-1>.
- Li, Z., Sun, J., Oh, S.-R., 2009. Path following for marine surface vessels with rudder and roll constraints: an MPC approach. In: 2009 American Control Conference. <https://doi.org/10.1109/acc.2009.5160302>. St. Louis, MO, USA.
- Martinsen, A.B., Lekkas, A.M., Gros, S., 2022. Reinforcement learning-based NMPC for tracking control of ASVs: theory and experiments. *Control Eng. Pract.* <https://doi.org/10.1016/j.conengprac.2021.105024>.
- Mockus, J., 1989. *Bayesian Approach to Global Optimization: Theory and Applications*. Oh, S.-R., Sun, J., 2010. Path following of underactuated marine surface vessels using line-of-sight based model predictive control. *Ocean Eng.* <https://doi.org/10.1016/j.oceaneng.2009.10.004>.
- Papadimitrakis, M., Stogiannos, M., Sarimveis, H., Alexandridis, A., 2021. Multi-ship control and collision avoidance using MPC and RBF-based trajectory predictions. *Sensors*. <https://doi.org/10.3390/s21216959>.
- PIANC, 1992. Capability of ship manoeuvring simulation models for approach channels and fairways in harbours. In: Report of Working Group No. 20 of Permanent Technical Committee II.
- Powell, M.J.D., 1994. *A Direct Search Optimization Method that Models the Objective and Constraint Functions by Linear Interpolation*. Springer.
- Putri, D.K.R., Subchan, Adzkiya, D., Asfihani, T., 2019. Ship heading control for dubins path tracking and collision avoidance using model predictive control. In: 2019 Int. Conf. Inf. Commun. Technol.. <https://doi.org/10.1109/icoiact46704.2019.8938455>.
- ROS2 [WWW Document]. <https://www.ros.org/>, 2022 (accessed 2.6.23).
- Sandeepkumar, R., Rajendran, S., Mohan, R., Pascoal, A.M., 2022. A unified ship manoeuvring model with a nonlinear model predictive controller for path following in regular waves. *Ocean Eng.* <https://doi.org/10.1016/j.oceaneng.2021.110165>.
- Shen, C., Buckham, B., Shi, Y., 2017. Modified C/GMRES algorithm for fast nonlinear model predictive tracking control of AUVs. *IEEE Trans. Control Syst. Technol.* <https://doi.org/10.1109/tcst.2016.2628803>.
- Shen, C., Shi, Y., Buckham, B., 2019. Path-following control of an AUV: a multiobjective model predictive control approach. *IEEE Trans. Control Syst. Technol.* <https://doi.org/10.1109/tcst.2018.2789440>.
- SIMMAN2020 [WWW Document]. https://simman2020.kr/contents/Ship_data_summary.php, 2023.
- Wang, W., Mateos, L.A., Mateos, L.A., Park, S., Leoni, P., Gheneti, B., Duarte, F., Duarte, F., Ratti, C., Rus, D., 2018. Design, modeling, and nonlinear model predictive tracking control of a novel autonomous surface vehicle. In: 2018 IEEE International Conference on Robotics and Automation (ICRA). <https://doi.org/10.1109/icra.2018.8460632>. Brisbane, QLD, Australia.
- Xu, P., Qin, H., Ma, J., Deng, Z., Xue, Y., 2023. Data-driven model predictive control for ships with Gaussian process. *Ocean Eng.* <https://doi.org/10.1016/j.oceaneng.2022.113420>.
- Xue, Y., Wang, X., Liu, Y., Xue, G., 2021. Real-time nonlinear model predictive control of unmanned surface vehicles for trajectory tracking and collision avoidance. In: International Conference on Mechatronics and Robotics Engineering, ICMRE 2021. Institute of Electrical and Electronics Engineers Inc., pp. 150–155. <https://doi.org/10.1109/ICMRE51691.2021.9384818>.
- Zhang, H., Zhang, X., Cao, T., Bu, R., 2022a. Active disturbance rejection control for ship path following with Euler method. *Ocean Eng.* 247, 110516 <https://doi.org/10.1016/j.oceaneng.2021.110516>.
- Zhang, J., Sun, T., Liu, Z., Liu, Z., 2017. Robust model predictive control for path-following of underactuated surface vessels with roll constraints. *Ocean Eng.* <https://doi.org/10.1016/j.oceaneng.2017.07.057>.
- Zhang, M., Hao, S., Wu, D., Chen, M.-L., Yuan, Z., 2022b. Time-optimal obstacle avoidance of autonomous ship based on nonlinear model predictive control. *Ocean Eng.* <https://doi.org/10.1016/j.oceaneng.2022.112591>.
- Zhang, R., Chen, Y., Sun, Z., Sun, F., Xu, H., 2000. Path control of a surface ship in restricted waters using sliding mode. *IEEE Trans. Control Syst. Technol.* 8, 722–732. <https://doi.org/10.1109/87.852916>.
- Zhao, Y., Zhang, Z., Wang, J., Wang, H., 2022. Fin-rudder joint control based on improved linear-quadratic-regulator algorithm. *IEEE Access* 10, 111105–111114.
- Zheng, H., Negenborn, R.R., Lodewijks, G., 2014. Trajectory tracking of autonomous vessels using model predictive control. *IFAC Proc.* <https://doi.org/10.3182/20140824-6-za-1003.00767>.
- Zheng, H., Zheng, H., Zheng, H., Wu, J., Wu, J., Wu, W., Zhang, Y., 2020. Robust dynamic positioning of autonomous surface vessels with tube-based model predictive control. *Ocean Eng.* <https://doi.org/10.1016/j.oceaneng.2019.106820>.